SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Xeno Force
Project / Product Name	ArcTalent
Final Live Deployment URL	https://arc-talent--emmanuelheng.replit.app/
Codebase Repository URL	https://github.com/MantouOP/ArcTalent
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2. Project Description

[Maximum 250 words]

ArcTalent helps working adults answer a difficult question: **"What should I learn now when nobody knows how quickly AI will change my career?"**

For people with limited time and money, choosing the wrong training pathway can mean months of effort without a useful next step. SMEs face a related challenge: understanding which existing employees could develop into new roles before resorting to external recruitment or redundancy.

ArcTalent turns a person's skills, interests, career goals, weekly learning time and budget into an explainable reskilling plan. It compares career options under moderate and rapid AI-adoption scenarios, identifies a shared foundation across the leading options, and shows where later learning steps diverge. Each scheduled course includes a direct provider link.

For SMEs, the workforce planner converts an entered report into an editable proposal covering team profiles, potential destinations, internal openings and training constraints. Managers can compare individual learning needs, grouped training opportunities and estimated costs.

Generative AI supports interpretation and guidance, while a transparent matching and scheduling engine calculates scores, gaps and plans. Users review suggestions and explicitly confirm skill-level changes.

The deployed prototype supports English, Bahasa Melayu and Simplified Chinese, PDF exports, study coaching and a learning track record. ArcTalent's contribution is a practical decision process: helping people take a useful next learning step while keeping their future options open.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

ArcTalent is a full-stack web application built with **Next.js, React and TypeScript**, deployed on **Replit**. Its architecture combines generative AI for interpreting user input with a shared, explainable engine for calculating recommendations.

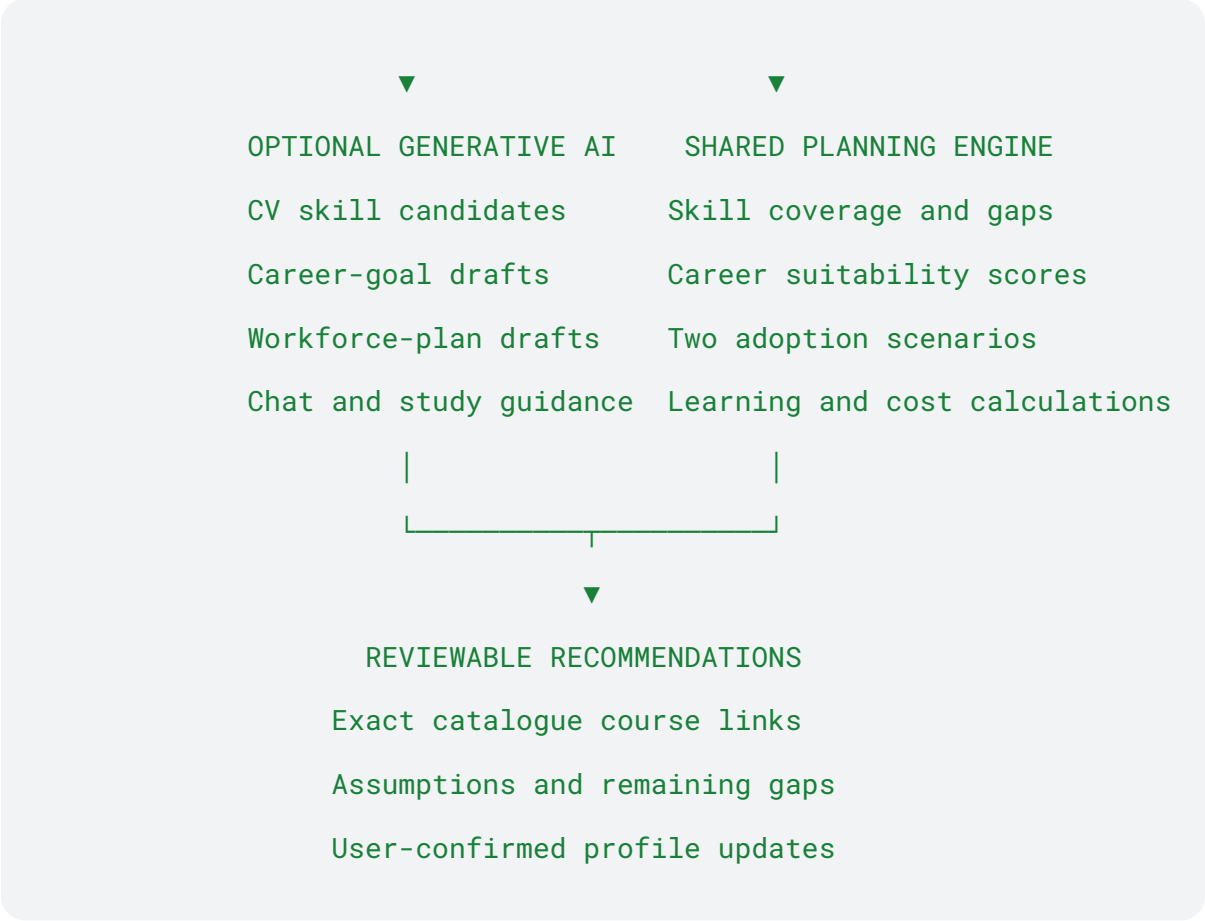
Application flow

```
None
INDIVIDUAL LEARNER                          SME MANAGER

Skills, interests, CV,                      Workforce report,

career goal, time and budget               team goals and constraints

             |                                        |
             └──────────────────┬─────────────────────┘
                                ▼
                    NEXT.JS / REACT INTERFACE
                Profile and career exploration
                Learning pathway and study coach
                Workforce planner and PDF exports
                                |
                                ▼
                          NEXT.JS API LAYER
                 Request and response validation
                 PDF extraction and report generation
                 AI requests and evidence refresh
                      |                    |
```



Frontend and interaction. The interface provides individual and SME workspaces. Users can enter their own information, inspect recommendations, compare scenarios and export plans. Detailed visualisations are available through expandable sections so the main workflow remains readable.

Backend and validation. Next.js route handlers process AI requests, text-based PDFs, evidence refreshes and PDF exports. Zod schemas validate structured inputs and outputs. AI-generated proposals are checked before they enter application state.

Generative AI integration. The application supports RunPod and an HTTPS OpenAI-compatible gateway through the Vercel AI SDK. AI can interpret CV text, suggest career requirements, draft workforce plans and provide learning guidance. Supported functions have labelled local fallbacks when AI is unavailable. Course destinations and quantitative outputs are derived from application data and calculations.

Data structures. The prototype stores typed JSON objects in browser localStorage rather than a cloud database. These include profiles, skill levels, interests, learning constraints, completed courses, workforce settings and planning preferences. Evidence snapshots retain their dates, source type and skill-demand values.

Human review. Suggested skills and workforce proposals remain editable. Learning activity and practice badges do not independently certify proficiency; users explicitly confirm changes to their skill levels.

Supporting evidence: Source repository · Verified release record

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

During Phase 2, we developed ArcTalent from a preliminary career-planning prototype into a more complete individual and SME reskilling workflow.

For individuals, we added custom skill and career inputs, course-linked pathways, study coaching, practice reflections and a per-profile learning track record. We simplified the interface so users can begin with their next decision and expand detailed analytics when needed.

For SMEs, we introduced a workforce planner that accepts a report and drafts team profiles, potential destinations, openings and budget assumptions. An editable redeployment table connects those proposals to calculated learning requirements, grouped training opportunities and PDF reports.

A significant integration problem arose when the AI gateway rejected open-ended maps in the workforce planner's structured-output schema. This caused requests to fall back to local parsing. We replaced those maps with explicit lists, retained application-side validation and added a regression test. The corrected live release successfully generated a proposal from a fictional eight-person retail workforce report.

Deployment also exposed Replit's image-size limit. We adjusted deployment exclusions for generated dependencies and build folders in inactive releases, preserving their source for rollback.

The verified release `0cb3c41` passed **250 unit tests, TypeScript checks, a production build and 39 production integration checks**. Its release process also verified that all **290 tracked files** matched the intended source before and after preparation.

Evidence: Release verification and problem-solving record

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

<table>
<tr><th>Metric Category</th><th>Your Response</th></tr>
<tr><td>Target SDG Metric Achievement
Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers
e.g., “Achieved a 24.5% reduction in computational execution overhead against the 20% initial target.”</td><td>ArcTalent’s primary contribution is to SDG 8: Decent Work and Economic Growth, supported by SDG 4: Quality Education and SDG 10: Reduced Inequalities. It aims to make workforce development more accessible and help employers explore internal development opportunities.

We evaluated the Phase 1 targets through a reproducible benchmark using eight fictional worker profiles, their learning constraints and a dated simulated evidence snapshot.

<table><tr><th>Phase 1 KPI and target</th><th>Baseline used for comparison</th><th>Final prototype result</th></tr><tr><td>At least 40% average skill-gap reduction</td><td>Each fictional worker’s initial gap against their recommended destination</td><td>Modelled average reduction of 79% under moderate adoption and 76% under rapid adoption. In both scenarios, 7 of 8 workers reached the 40% target.</td></tr></table></td></tr>
</table>

Metric Category	Your Response		
	At least 35% lower SME training cost	Individual commercial seats for the same scheduled learning, including paid study time	Under moderate adoption, the model estimated **RM15,269** for individual seats versus **RM4,364** for the grouped plan: approximately **71% lower**. Under rapid adoption, the grouped estimate was **RM3,564**, approximately **77% lower**.
	100% shared-foundation study during the first four weeks	Scheduled hours in the first four weeks	The average was **92%**. Five workers reached 100%; three completed their entire foundation earlier and then began role-specific study. The original numerical target was therefore **not fully met**, although foundation-first sequencing was maintained.
	100% human-confirmed adoption of suggested CV skills	No extracted candidate should automatically change the saved profile	The implementation requires review and saving before suggested skills enter the profile. This is an **enforced software control**, rather than a measured population outcome.
	The cost benchmark allocated **109 learner-hours to five workers** within the default company's capacity constraints. Its commercial-seat comparison uses an		

Metric Category	Your Response
	assumed RM125 per learner-hour, alongside trainer and paid-study-time assumptions. These figures demonstrate the prototype's planning behaviour. Skill gains are projected using course-to-skill assumptions, and cost savings are modelled comparisons rather than observed employer expenditure. We have not yet measured employment, wage, productivity or environmental outcomes. **Evidence:** Impact benchmark and assumptions
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	The benchmark is reproducible through `npm run impact`. It uses the application's matching, scheduling and cost-comparison functions rather than a separate spreadsheet calculation. For each fictional worker, the output records the selected route, initial skill-gap distance, projected remaining gap, scheduled hours, foundation allocation and course fees. It repeats the comparison under both AI-adoption scenarios. The model assumes that a scheduled course brings its mapped skills to working level, defined as level 2 of 3. This assumption allows consistent pathway comparison; it does not establish actual proficiency. Real-world validation would require independent assessment before and after learning. The cost model includes the same paid study time on both sides of the comparison and exposes its price assumptions for adjustment. The benchmark, source code and dated

Metric Category	Your Response
	deployment verification provide a traceable chain from input to calculation to displayed result.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	ArcTalent makes reskilling decisions more understandable for people who cannot commit to lengthy or expensive retraining. It helps users recognise existing strengths, identify a feasible next step and compare options under uncertainty. For SMEs, the application provides a structured starting point for workforce-development conversations. Managers can review potential internal moves and shared training needs with employees before making decisions. Partnerships with employers, universities, training providers and workforce organisations could turn this prototype into a practical support service. Its social value would be evaluated through participation, learning completion, independently assessed skill improvement and suitable internal opportunities—not by treating recommendation scores as employment outcomes.
Deployment & Commercial Feasibility	ArcTalent is accessible through a deployed web application with server-side APIs and PDF reporting. The

Metric Category	Your Response
e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.	prototype's browser-based storage keeps its initial infrastructure requirements small. A commercial version could offer an employer workspace for training-roadmap preparation and progress tracking, alongside institutional access for career or workforce-development programmes. Paid features could include organisation accounts, permissioned data integrations, reporting and support. Before commercial rollout, we would validate customer demand, delivery costs, data governance and local training relevance through a controlled pilot.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

A. Skill-gap calculation

For a destination role:

```
None
Missing skill levels

  = Σ max(required level - current level, 0)


Skill coverage

  = 1 - (missing skill levels / total required skill levels)


Skill-gap distance

  = 100 × (1 - skill coverage)


Projected gap reduction (%)

  = 100 × (initial distance - projected distance) / initial distance
```

For example, the fictional worker Daniel's gap distance changes from approximately **54.5 to 18.2**, giving a projected reduction of approximately **67%**.

The model applies the same calculation to all eight fictional workers and averages their individual percentage reductions. The resulting averages are **79% under moderate adoption** and **76% under rapid adoption**.

B. Training-cost calculation

The following logic is implemented in the cost-comparison module:

```
None
const studyTime = learnerHours * assumptions.wageRate;

const individual = seatFees + studyTime;

const grouped = planFees + trainers + studyTime;


const saving =

  individual > 0
```

```
    ? Math.round(1000 * (1 - grouped / individual)) / 10
    : null;
```

Both options include paid study time. The grouped option additionally includes catalogue fees and the modelled trainer cost for shared courses.

Using the rounded moderate-adoption benchmark:

```
Individual-seat cost = RM15,269
Grouped-plan cost    = RM4,364


Estimated saving
  = 100 × (1 - 4,364 / 15,269)
  ≈ 71%
```

C. Reproducible benchmark output

```
Moderate adoption:
Average projected skill-gap reduction: 79%
Workers reaching the 40% target: 7 of 8
Average foundation share in weeks 1–4: 92%


Rapid adoption:
Average projected skill-gap reduction: 76%
Workers reaching the 40% target: 7 of 8
Average foundation share in weeks 1–4: 92%
```

These calculations substantiate the **simulation results** reported above. Their real-world validity depends on testing the course-to-skill assumptions and cost inputs with learners and employers.

Evidence: Matching engine · Cost model · Impact benchmark

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Our initial market would be Malaysian SMEs with administrative, retail and service teams that need practical reskilling support but lack dedicated workforce-planning capacity.

With funding or industry support, we would begin with a proposed pilot involving three to five SMEs and approximately 40–60 workers. Over six to eight weeks, we would evaluate pathway relevance, completion, staff time, actual training expenditure and independently assessed skill improvement. This would test whether the prototype's projected benefits translate into useful outcomes.

We would seek permissioned Malaysian vacancy data and direct relationships with training providers to verify course availability, fees and HRD Corp eligibility. Potential collaboration through universities, workforce institutions or Rotary-linked employers could support recruitment and evaluation.

To process organisational data at scale, we would introduce authenticated accounts, a secured relational database, role-based access, audit records and retention controls. Background processing, caching and bounded AI requests would help manage workload and inference costs.

Our commercial hypothesis is an employer subscription for workforce planning, reporting and support, with institutional licences for education or workforce programmes. Pricing would follow evidence of customer value and delivery cost.

Expansion would be driven by demonstrated learning outcomes and employer demand, while keeping recommendations explainable and important decisions under human review.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

Name	Type	Licence / applicable terms
Next.js	Application framework	MIT
React	Interface library	MIT
TypeScript	Language and tooling	Apache License 2.0

Zod	Validation library	MIT
Vercel AI SDK and OpenAI-compatible provider package	AI integration libraries	Apache License 2.0
PDFKit	PDF generation library	MIT
pdf-parse	PDF processing library	Apache License 2.0
Lucide React	Icon library	ISC
Qwen2.5-7B-Instruct, used through RunPod where configured	Pre-trained model	Apache License 2.0
GPT-5.4 mini, used in the documented AI-gateway verification	Hosted model/API	Applicable provider and gateway terms
O*NET occupational references	Public occupational reference material	Applicable O*NET terms and attribution requirements
Arbeitnow job-posting feed	Public job data	Provider usage and attribution terms

arXiv research metadata	Public research metadata	arXiv API terms; descriptive metadata is CC0, while papers retain their own rights
External course-provider pages and linked videos	Educational references	Respective provider or creator terms; no transfer of ownership

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

[Signature]



[Date]

1.11.2026